Drinfeld's Reciprocity Law

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These notes reorganize the mathematical architecture of Drinfeld's original article into a continuous narrative. They begin with formal $\mathcal{O}$-modules, pass through Drinfeld modules, level structures, moduli, elliptic sheaves, rigid uniformization, and harmonic cochains, and culminate in the cohomological comparison that produces automorphic and Galois realizations on the same space. The endpoint is Drinfeld's proof of the Langlands correspondence for GL_2 over global function fields in the case where the component at infinity is special.

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1. Introduction

The purpose of these notes is to explain why Drinfeld's proof does not merely string together several ingenious constructions, but rather unfolds a single modular story. The story starts locally. One first studies one-dimensional formal groups equipped with an action of the ring of integers of a local function field. Such objects are called formal $\mathcal{O}$-modules. They provide the infinitesimal language in which Frobenius, torsion, height, and deformation are visible without any global geometry at all.

The local picture then acquires a global envelope. Once the ground ring is replaced by the affine ring

$$A = \Gamma(X \setminus \{\infty\}, \mathcal{O}_X)$$

of a smooth projective curve over $\mathbb{F}_q$, the same Frobenius-polynomial formalism can be interpreted as an action of A on the additive group scheme $\mathbb{G}_a$. This global object is a Drinfeld module. Level structures rigidify automorphisms, so the resulting moduli problems become representable by algebraic varieties. In rank two these are the Drinfeld modular curves whose cohomology eventually carries both Hecke operators and a Galois action.

There is also a geometric reformulation, and it is indispensable. The correct geometric avatars of Drinfeld modules are *elliptic sheaves*. On the one hand they encode the same algebraic data. On the other hand they place the theory inside the language of vector bundles, Frobenius modifications, and linear series on the complete curve X. This is the point where Riemann–Roch enters naturally: the pole filtration at ∞, the dimensions of spaces of sections, and the periodicity conditions are all controlled by the same cohomological bookkeeping.

The analytic phase begins when one studies the rigid analytification of the modular varieties. The Drinfeld upper half-plane

$$\Omega^2 = \mathbb{P}^1(\mathbb{C}_\infty) \setminus \mathbb{P}^1(K_\infty)$$

uniformizes the rank-two modular curve. Its admissible covering is governed by the Bruhat–Tits tree of $\mathrm{PGL}_2(K_\infty)$. From this one obtains a remarkably concrete cohomological description: the relevant degree-one cohomology is identified with harmonic cochains on the tree. Harmonicity is not a decorative condition; it is the exact algebraic expression of the cocycle relation imposed by the rigid covering.

The automorphic meaning then emerges almost by force. Harmonic cochains realize the special representation at infinity, so the cohomology of the modular curve becomes an adelic automorphic module. The commuting action of Hecke operators away from ∞ and the Galois group of K on the same cohomology space produces the desired Galois representation attached to an automorphic representation. This is the function-field analogue of the Eichler–Shimura philosophy.

The final statement we aim at is the following.

Theorem 1.1 (Drinfeld reciprocity law, global form)**.** *Let K be a global function field, let ∞ be a fixed place of K, and let $\ell \neq \mathrm{char}(K)$. Consider:*

1. *the set Π_∞ of irreducible cuspidal automorphic representations π of $\mathrm{GL}_2(\mathbb{A}_K)$ whose local component at ∞ is the special representation;*
2. *the set Σ_∞ of continuous irreducible 2-dimensional ℓ-adic representations*
$$\sigma : \mathrm{Gal}(K^{\mathrm{sep}}/K) \longrightarrow \mathrm{GL}_2(\mathbb{Q}_\ell)$$
which are unramified outside finitely many places and satisfy the local condition corresponding to the special representation at ∞.

Then there is a natural bijection
$$\Pi_\infty \xrightarrow{\sim} \Sigma_\infty$$
compatible with unramified local parameters, central characters, and twisting by characters.

The deepest representation-theoretic comparison theorems belong to Drinfeld's article and its descendants. In these notes we expand every section as far as possible within a self-contained exposition. For the algebraic, local, geometric, and cohomological steps we give full proofs; for the final global matching we isolate the remaining inputs with precise reductions so that the logical structure of the proof becomes transparent.

2. Arithmetic framework and notation

Let X be a smooth, projective, geometrically irreducible curve over $\mathbb{F}_q$. Fix a closed point $\infty \in |X|$ and write
$$K = \mathbb{F}_q(X), \qquad A = \Gamma(X \setminus \{\infty\}, \mathcal{O}_X).$$
Then A is a Dedekind domain with field of fractions K. The place ∞ determines a valuation v_∞, a completion K_∞, its valuation ring $\mathcal{O}_\infty$, and a completed algebraic closure $\mathbb{C}_\infty$.

For every finite place $x \neq \infty$ let K_x be the completion of K at x, let $\mathcal{O}_x$ be its ring of integers, let ϖ_x be a uniformizer, and let $\kappa(x)$ be its residue field. We denote by $\mathbb{A}_K$ the adèle ring of K and by $\mathbb{A}^\infty$ the ring of finite adèles away from ∞.

The degree at infinity is normalized by

$$\deg_\infty(a) = -\deg(\infty)\, v_\infty(a), \qquad a \in K^\times,$$

so that

$$|a|_\infty = q^{\deg_\infty(a)}.$$

For $a \in A \setminus \{0\}$ one has

$$\#(A/aA) = q^{\deg_\infty(a)}.$$

Example 2.1. The basic model is

$$X = \mathbb{P}^1_{\mathbb{F}_q}, \qquad \infty = [1:0].$$

Then

$$A = \mathbb{F}_q[T], \qquad K = \mathbb{F}_q(T), \qquad K_\infty = \mathbb{F}_q((T^{-1})).$$

Most formulas become completely explicit in this case, yet the conceptual structure is already the general one.

Two simple algebraic facts about A will be used repeatedly.

Proposition 2.2. *The ring A is a Dedekind domain. Every nonzero ideal $I \subset A$ factors uniquely as a product of prime ideals, and for every nonzero ideal I there exists an ideal J such that IJ is principal and $I + J = A$.*

Proof. Since $X \setminus \{\infty\}$ is a smooth affine curve, its coordinate ring A is Noetherian, integrally closed, and of Krull dimension one. This is exactly the defining package for a Dedekind domain.

For the second assertion, write $I = \prod_i \mathfrak{p}_i^{n_i}$. By weak approximation on the curve, or equivalently by the surjectivity of the divisor map modulo principal divisors with prescribed support, choose $a \in K^\times$ with

$$v_{\mathfrak{p}_i}(a) = n_i \quad \text{for all } i$$

and $v_{\mathfrak{q}}(a) \geq 0$ for every prime $\mathfrak{q} \neq \mathfrak{p}_i$. Then $(a) = IJ^{-1}$ for some fractional ideal J^{-1}, hence $IJ = (a)$ for an integral ideal J. Multiplying a by a suitable element of A not vanishing at any $\mathfrak{p}_i$ if necessary, one may arrange $I + J = A$. In ideal-theoretic language this just says that every ideal class contains infinitely many ideals prime to a given finite set of places. $\square$

Proposition 2.3 (Chinese remainder in the present setting)**.** *Let $I, J \subset A$ be coprime nonzero ideals. Then the canonical map*

$$A/(IJ) \longrightarrow A/I \times A/J$$

is an isomorphism. Consequently, for every $d \geq 1$,

$$(A/IJ)^d \cong (A/I)^d \times (A/J)^d.$$

Proof. Because $I + J = A$, there exist $u \in I$ and $v \in J$ with $u + v = 1$. The standard Chinese remainder proof applies verbatim: surjectivity follows because

$$(a, b) \longmapsto av + bu \pmod{IJ}$$

maps to (a, b), while injectivity follows from $I \cap J = IJ$ in a Dedekind domain for coprime ideals. The statement for d-fold products is immediate by taking Cartesian powers. $\square$

3. Formal $\mathcal{O}$-modules

3.1. Additive operators and Frobenius polynomials

Fix a complete discrete valuation ring $\mathcal{O}$ of characteristic $p > 0$ with residue field $\mathbb{F}_q$ and uniformizer ϖ. Let R be a complete separated $\mathcal{O}$-algebra for some ideal of definition.

The additive formal group over R is

$$\widehat{\mathbb{G}}_{a,R}(X,Y) = X + Y.$$

In characteristic p the continuous endomorphisms of $\widehat{\mathbb{G}}_{a,R}$ are exactly the additive power series

$$P(T) = \sum_{i \geq 0} b_i T^{q^i}, \qquad b_i \in R.$$

It is useful to encode them in the skew power-series ring

$$R\{\{\tau\}\} = \left\{ \sum_{i \geq 0} b_i \tau^i \right\}, \qquad \tau b = b^q \tau.$$

Lemma 3.1. *Composition of additive series corresponds to multiplication in $R\{\{\tau\}\}$. More precisely, if*

$$P(T) = \sum_i a_i T^{q^i}, \qquad Q(T) = \sum_j b_j T^{q^j},$$

then

$$P \circ Q(T) = \sum_{i,j} a_i b_j^{q^i} T^{q^{i+j}},$$

which is the product of $\sum_i a_i \tau^i$ and $\sum_j b_j \tau^j$ in the skew ring.

Proof. We simply expand:

$$P(Q(T)) = \sum_i a_i \left(\sum_j b_j T^{q^j} \right)^{q^i} = \sum_i a_i \sum_j b_j^{q^i} T^{q^{i+j}}.$$

The defining commutation rule $\tau b = b^q \tau$ yields exactly the same coefficient formula for multiplication in $R\{\{\tau\}\}$. □

Definition 3.2. A one-dimensional formal $\mathcal{O}$-module over R is a one-dimensional commutative formal group F over R together with a ring homomorphism

$$[a]_F : \mathcal{O} \longrightarrow \mathrm{End}(F)$$

such that the linear term of $[a]_F(T)$ is the image of a in R.

When F is identified with $\widehat{\mathbb{G}}_{a,R}$, this means that each $[a]_F(T)$ is an additive series and

$$[a]_F(T) \equiv aT \pmod{T^2}.$$

This tangent normalization is the local shadow of the defining normalization for Drinfeld modules.

Example 3.3 (A Lubin–Tate style model in equal characteristic)**.** Take $\mathcal{O} = \mathbb{F}_q[[\varpi]]$ and $R = \mathcal{O}$. On $\widehat{\mathbb{G}}_{a,\mathcal{O}}$ define

$$[\varpi]_F(T) = \varpi T + T^q, \qquad [a]_F(T) = aT \text{ for } a \in \mathbb{F}_q.$$

There is a unique extension to an $\mathcal{O}$-action, and reducing modulo ϖ gives

$$[\varpi]_{\bar{F}}(T) = T^q,$$

so the special fiber has height 1. This is the local rank-one prototype that reappears globally in the Carlitz module and, after completion at a finite place, in every rank-one Drinfeld module.

3.2. Strict coordinates, normalized $[\varpi]$-series, and local moduli

A change of coordinate on a one-dimensional formal group is a power series

$$u(T) = u_1 T + u_2 T^2 + \cdots \in R[\![T]\!], \qquad u_1 \in R^\times,$$

with composition inverse. It is called *strict* if $u_1 = 1$. Conjugating the $\mathcal{O}$-action by such a series changes coordinates without changing the underlying formal $\mathcal{O}$-module.

Lemma 3.4. *Let F and G be one-dimensional formal $\mathcal{O}$-modules over an $\mathbb{F}_q$-algebra R, written in coordinates on $\widehat{\mathbb{G}}_a$. A strict isomorphism $u : F \to G$ is determined by the identity*

$$u \circ [\varpi]_F = [\varpi]_G \circ \nu.$$

In other words, once the $\mathbb{F}_q$-linear structure is fixed, compatibility with the action of the single topological generator ϖ forces compatibility with all of $\mathcal{O} = \mathbb{F}_q[[\varpi]]$.

Proof. The two ring homomorphisms

$$a \longmapsto \nu \circ [a]_F \circ \nu^{-1}, \qquad a \longmapsto [a]_G$$

from $\mathcal{O}$ to $\mathrm{End}(\widehat{\mathbb{G}}_{a,R})$ agree on $\mathbb{F}_q$ because both actions have the same tangent normalization and are $\mathbb{F}_q$-linear. They also agree on ϖ by hypothesis. Since $\mathcal{O}$ is the ϖ-adic completion of $\mathbb{F}_q[\varpi]$, every element of $\mathcal{O}$ is a limit of polynomials in ϖ with coefficients in $\mathbb{F}_q$. Composition in $R\{\{\tau\}\}$ is continuous for the (ϖ, T)-adic topology, so the two homomorphisms agree on all limits as well. Thus they agree on every $a \in \mathcal{O}$. □

Proposition 3.5. *Let k be a perfect field of characteristic p equipped with the special $\mathcal{O}$-characteristic $\mathcal{O} \twoheadrightarrow k$. Let F be a one-dimensional formal $\mathcal{O}$-module of finite height h over k and write*

$$[\varpi]_F(T) = c_h T^{q^h} + \sum_{i>h} c_i T^{q^i}, \qquad c_h \in k^\times.$$

Then there exists a linear change of coordinate $u(T) = \alpha T$ with $\alpha \in k^\times$ such that, in the new coordinate,

$$[\varpi]_F(T) = T^{q^h} + \sum_{i>h} c_i' T^{q^i}.$$

In particular, every height-h formal $\mathcal{O}$-module begins, after a strict normalization of its tangent and a linear rescaling, with the pure q^h-Frobenius term.

Proof. Conjugating by $u(T) = \alpha T$ replaces $[\varpi]_F$ by

$$u \circ [\varpi]_F \circ v^{-1}(T) = \alpha\,[\varpi]_F(\alpha^{-1}T) = \alpha c_h \alpha^{-q^h} T^{q^h} + \sum_{i>h} \alpha c_i \alpha^{-q^i} T^{q^i}.$$

So the new leading coefficient is $c_h\alpha^{1-q^h}$. Because k is perfect, the equation $\alpha^{q^h-1} = c_h$ has a solution in $k^\times$. For such an α the leading coefficient becomes 1. The remaining coefficients are simply the rescaled values $c_i' = \alpha^{1-q^i}c_i$. □

Remark 3.6. This elementary normalization is the first glimpse of the Honda–Lubin–Tate philosophy. Finite height means that the ϖ-series starts with a pure Frobenius monomial. All higher terms measure deformation away from that monomial, and those higher coefficients are precisely what later become local moduli coordinates.

3.3. Height

Assume now that k is a field on which the uniformizer ϖ acts by zero. Then

$$[\varpi]_F(T) = \sum_{i\geq 1} c_i T^{q^i}$$

has no linear term. The first nonzero Frobenius term measures the singularity of the formal module.

Definition 3.7. The height of a one-dimensional formal $\mathcal{O}$-module F over k is the least $h \in \{1, 2, \dots\} \cup \{\infty\}$ such that

$$[\varpi]_F(T) = uT^{q^h} + \text{higher } q\text{-powers}, \qquad u \in k^\times.$$

Proposition 3.8. *Let F be a one-dimensional formal $\mathcal{O}$-module of finite height h over an algebraically closed field k of characteristic p. Then for every $n \geq 1$ the finite group scheme $F[\varpi^n]$ has rank q^{nh}.*

Proof. Write

$$[\varpi]_F(T) = uT^{q^h} + \sum_{i>h} c_i T^{q^i}, \qquad u \in k^\times.$$

Because composition of additive polynomials multiplies q-degrees, the first nonzero term of $[\varpi^n]_F$ is

$$u_n T^{q^{nh}}$$

for some $u_n \in k^\times$. Thus the additive polynomial $[\varpi^n]_F(T)$ has degree q^{nh} as an ordinary polynomial in T.

Over an algebraically closed field, the kernel of an additive polynomial of degree q^m defines a finite group scheme of rank q^m; indeed the coordinate algebra is $k[T]/([\varpi^n]_F(T))$, whose dimension as a k-vector space equals the polynomial degree because the leading coefficient is invertible. Therefore $\operatorname{rk} F[\varpi^n] = q^{nh}$. □

Corollary 3.9. *The integer h is the unique number such that* $\operatorname{rk} F[\varpi^n] = q^{nh}$ *for all $n \geq 1$.*

Proof. The proposition gives existence. Uniqueness follows because the rank sequence determines h from the first term:

$$q^h = \operatorname{rk} F[\varpi].$$

$\square$

Proposition 3.10 (Weierstrass form of torsion over a complete local base)**.** *Let R be a complete local $\mathcal{O}$-algebra with maximal ideal $\mathfrak{m}$, and let F be a one-dimensional formal $\mathcal{O}$-module over R whose reduction F_0 over $k = R/\mathfrak{m}$ has finite height h. Then for every $n \geq 1$ the power series $[\varpi^n]_F(T)$ is a distinguished polynomial of degree q^{nh} times a unit in $R[\![T]\!]$. Consequently*

$$R[\![T]\!]/([\varpi^n]_F(T))$$

is finite free of rank q^{nh} over R, and $F[\varpi^n]$ is finite flat of that rank.

Proof. Reduce first modulo $\mathfrak{m}$. By the preceding proposition, after a linear normalization of the special fiber we may write

$$[\varpi]_{F_0}(T) = T^{q^h} + \sum_{i>h} \bar{c}_i T^{q^i}.$$

Hence $[\varpi^n]_{F_0}(T)$ is an additive polynomial whose lowest nonzero q-power term is $T^{q^{nh}}$. Equivalently, the reduction of $[\varpi^n]_F(T)$ modulo $\mathfrak{m}$ is a polynomial of exact degree q^{nh} with leading coefficient in $k^\times$.

Write $[\varpi^n]_F(T) = \sum_{m \geq 0} a_m T^m$. The coefficient $a_{q^{nh}}$ is therefore a unit of R, and every coefficient a_m with $m < q^{nh}$ lies in $\mathfrak{m}$ unless $m = 1$, in which case the coefficient is ϖ^n. Since $\varpi \in \mathfrak{m}$, all coefficients below degree q^{nh} are topologically nilpotent. By the Weierstrass preparation theorem there exists a unique factorization

$$[\varpi^n]_F(T) = P_n(T)\, U_n(T),$$

where $U_n(T) \in R[\![T]\!]^\times$ and $P_n(T) \in R[T]$ is monic of degree q^{nh} with all lower coefficients in $\mathfrak{m}$. Therefore

$$R[\![T]\!]/([\varpi^n]_F(T)) \cong R[T]/(P_n(T)),$$

and the latter is finite free of rank q^{nh} with basis $1, T, \ldots, T^{q^{nh}-1}$. This is exactly the coordinate ring of $F[\varpi^n]$. $\square$

Theorem 3.11 (Lubin–Tate–Drinfeld local deformation space)**.** *Fix an algebraically closed field k of characteristic p and a height-h one-dimensional formal $\mathcal{O}$-module F_0 over k. The functor*

$$\mathrm{Def}_{F_0} : \{\textit{complete local } \mathcal{O}\textit{-algebras with residue field } k\} \longrightarrow \{\textit{sets}\}$$

that assigns to R the set of isomorphism classes of deformations of F_0 to R is pro-represented by a formal power series ring in $h-1$ variables:

$$R_{F_0}^{\mathrm{univ}} \cong \mathcal{O}[\![u_1, \ldots, u_{h-1}]\!].$$

After choosing a coordinate on the universal deformation, one may arrange that

$$[\varpi]_{\mathrm{univ}}(T) = \varpi T + u_1 T^q + \cdots + u_{h-1} T^{q^{h-1}} + T^{q^h}.$$

Proof. We indicate the standard recursive proof. By the previous normalization, every deformation may be written in some coordinate as

$$[\varpi]_F(T) = \varpi T + a_1 T^q + \cdots + a_{h-1} T^{q^{h-1}} + T^{q^h} + \sum_{i>h} a_i T^{q^i}.$$

A strict coordinate change $u(T) = T + b_1 T^q + b_2 T^{q^2} + \cdots$ acts on these coefficients triangularly: the coefficient of T^{q^m} in

$$\nu \circ [\varpi]_F \circ \nu^{-1}$$

depends only on $a_1, \ldots, a_m$ and on $b_1, \ldots, b_{m-1}$. One therefore kills the coefficients a_i for $i > h$ successively, solving at each stage a linear equation in the next unknown b_j. What remains is exactly a series of the displayed form.

Conversely, given arbitrary parameters $u_1, \ldots, u_{h-1}$ in a complete local $\mathcal{O}$-algebra R, the displayed formula defines an additive polynomial with special fiber of height h; by the lemma on strict isomorphisms, this determines an $\mathcal{O}$-action and hence a deformation of F_0. The same triangular calculation shows that any two such deformations are strictly isomorphic only when their parameter tuples coincide. Thus the deformation functor is represented by the ring $\mathcal{O}[\![u_1, \ldots, u_{h-1}]\!]$. □

Remark 3.12. For $h = 1$ no deformation parameter survives. This rigidity is the local reason Lubin–Tate theory produces explicit class field theory from a single formal module. For $h > 1$ the deformation space acquires genuine dimension, and these higher-dimensional local moduli are the prototype for the much larger shtuka spaces appearing later in the Langlands story.

3.4. Why formal $\mathcal{O}$-modules are the local beginning of the theory

A Drinfeld module will be a global action of A on $\mathbb{G}_a$ by Frobenius polynomials. If we complete the underlying group scheme at the identity and then complete A at a finite place x, we obtain a formal $\mathcal{O}_x$-module. The height controls the connected part of local torsion, just as the rank of the Drinfeld module controls the size of its global torsion. Local and global are therefore not separate worlds: the formal module is what one sees when the global object is magnified near one place and near the origin simultaneously.

4. Drinfeld modules

4.1. Definition over fields

Let k be a field of characteristic p together with an $\mathbb{F}_q$-algebra map

$$\gamma : A \longrightarrow k.$$

Write $k\{\tau\}$ for the skew polynomial ring in the Frobenius operator $\tau(r) = r^q$.

Definition 4.1. A Drinfeld A-module over k is a ring homomorphism

$$\phi : A \longrightarrow k\{\tau\}$$

such that:

1. $\phi(A) \not\subset k$;
2. if $D : k\{\tau\} \to k$ denotes projection to the constant term, then

$$D(\phi_a) = \gamma(a) \qquad \text{for all } a \in A.$$

The second condition says that the tangent action is the given structural action of A on k. The first excludes the degenerate case where no genuine Frobenius terms occur.

Example 4.2 (The Carlitz module)**.** For $A = \mathbb{F}_q[T]$ and a structural map $\gamma : A \to k$, the assignment

$$\phi_T = \gamma(T) + \tau$$

extends uniquely to a Drinfeld module. This is the rank-one Carlitz module.

Example 4.3 (A rank-two module over the polynomial ring)**.** Let $A = \mathbb{F}_q[T]$, let k be an $\mathbb{F}_q$-field, and choose $\theta, c, \Delta \in k$ with $\Delta \neq 0$. Then

$$\phi_T = \theta + c\tau + \Delta\tau^2$$

defines a rank-two Drinfeld module. The nonzero coefficient of τ^2 forces the q^2-term in every sufficiently large ϕ_a, hence the rank is two, while the ratio

$$j(\phi) = \frac{c^{q+1}}{\Delta}$$

plays the role of a modular parameter. This is the function-field analogue of writing an elliptic curve in a one-parameter normal form.

4.2. Morphisms and injectivity

Definition 4.4. If ϕ, ψ are Drinfeld modules over k, a morphism $u : \phi \to \psi$ is an element $u \in k\{\tau\}$ such that

$$u\phi_a = \psi_a u \qquad \text{for all } a \in A.$$

A nonzero morphism is called an isogeny.

Proposition 4.5. *Every Drinfeld module $\phi : A \to k\{\tau\}$ is injective.*

Proof. The ring $k\{\tau\}$ has no zero divisors. Hence $\ker(\phi)$ is a prime ideal of the Dedekind domain A. Suppose $\ker(\phi) \neq 0$. Then $\ker(\phi)$ is maximal, so the image $\phi(A)$ is a field. We claim that every invertible element of $k\{\tau\}$ lies in $k^\times$. Indeed, if

$$u = a_0 + a_1\tau + \cdots + a_n\tau^n$$

with $n > 0$, then for every nonzero $v \in k\{\tau\}$ one has

$$\deg_\tau(uv) = \deg_\tau(u) + \deg_\tau(v) > 0,$$

so $uv \neq 1$. Therefore only nonzero constants are units. Since a field inside $k\{\tau\}$ consists entirely of units except for zero, the image $\phi(A)$ must lie in k, contradicting the defining nondegeneracy condition. Thus $\ker(\phi) = 0$. $\square$

4.3. Rank

For $P = \sum_{i=0}^{n} a_i \tau^i \in k\{\tau\}$ with $a_n \neq 0$ define

$$\deg_\tau(P) = q^n,$$

and $\deg_\tau(0) = 0$. The normalization by q^n rather than n is convenient because composition multiplies these degrees.

Lemma 4.6. *For nonzero $P, Q \in k\{\tau\}$,*

$$\deg_\tau(PQ) = \deg_\tau(P)\deg_\tau(Q), \qquad \deg_\tau(P+Q) \le \max(\deg_\tau(P), \deg_\tau(Q)).$$

Proof. If

$$P = \sum_{i=0}^{m} a_i \tau^i, \qquad Q = \sum_{j=0}^{n} b_j \tau^j$$

with $a_m, b_n \neq 0$, then the highest τ-term of PQ is

$$a_m b_n^{q^m} \tau^{m+n},$$

which is nonzero. Hence $\deg_\tau(PQ) = q^{m+n} = q^m q^n$. The inequality for sums is obvious from the definition. □

Proposition 4.7. *There exists a unique real number $d > 0$ such that*

$$\deg_\tau(\phi_a) = q^{d \deg_\infty(a)} \qquad \textit{for all } a \in A \setminus \{0\}.$$

Moreover d is an integer. It is called the rank of ϕ.

Proof. Set

$$\mu(a) = \log_q \deg_\tau(\phi_a) \in \mathbb{Z}_{\geq 0}.$$

Because $\phi_{ab} = \phi_a \phi_b$, the preceding lemma shows

$$\mu(ab) = \mu(a) + \mu(b).$$

Similarly, from $\phi_{a+b} = \phi_a + \phi_b$ we obtain

$$\mu(a+b) \le \max(\mu(a), \mu(b)).$$

Thus

$$|a|_\phi := q^{\mu(a)}$$

defines a nonarchimedean absolute value on A, hence on K by

$$|a/b|_\phi = |a|_\phi / |b|_\phi.$$

Any nontrivial nonarchimedean absolute value on the global field K which is trivial on $\mathbb{F}_q^\times$ is a positive real power of $|\cdot|_\infty$ when it is nonnegative on A. Therefore there exists $d > 0$ such that

$$|a|_\phi = |a|_\infty^d = q^{d \deg_\infty(a)}.$$

Equivalently,

$$\mu(a) = d \deg_\infty(a).$$

It remains to show that d is an integer. Choose a nonconstant $a \in A$. Then $\deg_\tau(\phi_a) = q^{d \deg_\infty(a)}$ is an integral power of q, so $d \deg_\infty(a) \in \mathbb{Z}$. Since the integers $\deg_\infty(a)$ for $a \in A \setminus \mathbb{F}_q$ generate the subgroup $\deg(\infty)\mathbb{Z}$ of $\mathbb{Z}$, the denominator of d divides all such degrees. A standard ideal-theoretic argument using prime ideals of degree one after finite constant extension then shows $d \in \mathbb{Z}$. Concretely, one can pass to a finite extension of constants over which there exists a with $\deg_\infty(a) = 1$; rank is invariant under constant extension, so $d \in \mathbb{Z}$ already over the original field. □

Remark 4.8. In the standard case $A = \mathbb{F}_q[T]$ a rank-d Drinfeld module is uniquely determined by

$$\phi_T = \gamma(T) + c_1\tau + \cdots + c_d\tau^d, \qquad c_d \neq 0.$$

The rank is literally the highest Frobenius exponent appearing in ϕ_T.

4.4. Drinfeld modules over schemes

Definition 4.9. Let S be an $\mathbb{F}_q$-scheme over A. A Drinfeld module of rank d over S is a line bundle L over S together with a ring homomorphism

$$\phi : A \longrightarrow \mathrm{End}_{\mathbb{F}_q}(L)$$

such that Zariski locally on S, after choosing a trivialization $L \simeq \mathbb{G}_{a,S}$, each ϕ_a is given by an additive polynomial

$$\phi_a = \sum_{i=0}^{d \deg_\infty(a)} b_i(a)\tau^i$$

with $b_0(a)$ equal to the structural image of a and $b_{d \deg_\infty(a)}(a)$ invertible.

The point of this formulation is that the additive group need not be globally trivial. The scheme-theoretic version is the right one for moduli.

5. The Carlitz module and explicit class field theory

The rank-one theory deserves to be isolated before one moves into higher-rank moduli. It is the place where the Frobenius-polynomial formalism already produces a nontrivial reciprocity law, but in a form so explicit that one can see the mechanism by hand. The Carlitz module is therefore not merely the first example of a Drinfeld module. It is the function-field analogue of the multiplicative group in cyclotomic class field theory.

Throughout this section we specialize to the basic case

$$A = \mathbb{F}_q[T], \qquad K = \mathbb{F}_q(T).$$

5.1. The Carlitz module and its exponential

Definition 5.1. The *Carlitz module* is the rank-one Drinfeld module

$$C : A \longrightarrow K\{\tau\}, \qquad C_T = T + \tau.$$

For a polynomial $a(T) \in A$ the endomorphism C_a is obtained by evaluating a at $T + \tau$ inside the noncommutative ring $K\{\tau\}$.

Proposition 5.2. *The assignment $T \mapsto T + \tau$ extends uniquely to a rank-one Drinfeld module. For every nonzero $a \in A$ one has*

$$\deg_\tau C_a = q^{\deg a}.$$

Proof. Uniqueness is formal because $A = \mathbb{F}_q[T]$ is generated by T as an $\mathbb{F}_q$-algebra. For existence one defines $C_a = a(T + \tau)$ and checks directly that

$$C_{a+b} = C_a + C_b, \qquad C_{ab} = C_a \circ C_b.$$

Since $\tau T = T^q \tau$, the highest τ-power appearing in C_{T^n} is τ^n with nonzero coefficient, hence $\deg_\tau C_{T^n} = q^n$. For a general polynomial a of degree n, the leading term of $a(T + \tau)$ is still a nonzero multiple of τ^n, so the same formula holds. The tangent term is $D(C_a) = a(T)$, and therefore C is a Drinfeld module. The degree formula shows that the rank is 1. □

The next step is the analogue of the complex exponential for the multiplicative group. In the Carlitz case one can write it by a convergent $\mathbb{F}_q$-linear power series

$$e_C(z) = \sum_{i \geq 0} \frac{z^{q^i}}{D_i}, \qquad D_0 = 1,$$

where the denominators $D_i \in A$ are determined recursively by the functional equation.

Proposition 5.3. *There is a unique entire $\mathbb{F}_q$-linear function*

$$e_C(z) = z + \sum_{i \geq 1} c_i z^{q^i}$$

such that

$$e_C(Tz) = C_T(e_C(z)) = Te_C(z) + e_C(z)^q.$$

Its kernel is a free rank-one A-module generated by a nonzero period $\widetilde{\pi}_C \in \mathbb{C}_\infty$.

Proof. Write

$$e_C(z) = \sum_{i \geq 0} c_i z^{q^i}, \qquad c_0 = 1.$$

Comparing coefficients in the identity

$$e_C(Tz) = Te_C(z) + e_C(z)^q$$

gives the recursion

$$(T^{q^i} - T)c_i = c_{i-1}^q \qquad (i \geq 1).$$

Since $T^{q^i} - T \neq 0$, the coefficients are uniquely determined. Their valuations tend to $+\infty$, so the series converges on all of $\mathbb{C}_\infty$. The functional equation implies that the kernel is an A-submodule. As in the general uniformization theorem, the derivative at the origin is 1, so e_C is an isometry on a sufficiently small disk; hence the kernel is discrete. The rank computation follows by comparing kernels of C_a with $a^{-1}\Lambda/\Lambda$, where $\Lambda = \ker(e_C)$. Because C has rank one, Λ must be a free rank-one A-module. Choosing a generator defines the Carlitz period $\tilde{\pi}_C$. □

Remark 5.4. The formal similarity with the complex exponential is real but should not be overstated. The classical exponential linearizes multiplication on $\mathbb{G}_m$; the Carlitz exponential linearizes the Frobenius-polynomial action of A on $\mathbb{G}_a$. The underlying one-dimensional group is additive, yet the reciprocity law it controls is cyclotomic in spirit.

5.2. Torsion and the finite-level reciprocity law

For a nonzero ideal $I = (a) \subset A$ the I-torsion of the Carlitz module is

$$C[I] = \{x \in \mathbb{C}_\infty \mid C_a(x) = 0\}.$$

Because the module has rank one, this is a free A/I-module of rank one.

Proposition 5.5. *For every nonzero $a \in A$ the polynomial $C_a(X)$ has exactly $q^{\deg a}$ roots in $\mathbb{C}_\infty$, all simple when a is prime to the characteristic of the base. Hence*

$$C[a] \cong A/aA$$

as A-modules.

Proof. The derivative of $C_a(X)$ is its constant term a, so over characteristic prime to a the roots are simple. The degree formula of the preceding proposition shows that C_a has ordinary polynomial degree $q^{\deg a}$. Since e_C uniformizes C, its kernel comparison gives

$$C[a] \cong a^{-1}A/A.$$

As an A-module the right-hand side is canonically isomorphic to A/aA by multiplication by a. □

Let $K(C[a])$ denote the field generated over K by the a-torsion points.

Proposition 5.6. *The natural Galois action on $C[a]$ defines a homomorphism*

$$\rho_a : \mathrm{Gal}(K^{\mathrm{sep}}/K) \longrightarrow (A/aA)^\times.$$

For every finite place $\mathfrak{p} \nmid a$, the arithmetic Frobenius at $\mathfrak{p}$ acts on $C[a]$ by the class of the monic generator of $\mathfrak{p}$ modulo a.

Proof. Every Galois automorphism commutes with the coefficients of C_a, hence preserves the A-module $C[a]$. After choosing an A/aA-basis of the free rank-one module $C[a]$, the action is given by multiplication by an invertible class, which defines ρ_a.

For the Frobenius statement, let $\lambda \in C[a]$ and reduce modulo a prime $\mathfrak{p}$ not dividing a. Since the coefficients of C_b lie in A, one has

$$\lambda^{q^{\deg \mathfrak{p}}} \equiv C_{\pi_{\mathfrak{p}}}(\lambda) \pmod{\mathfrak{p}}$$

where $\pi_{\mathfrak{p}}$ is the monic generator of $\mathfrak{p}$. Indeed, both sides are roots of the same additive polynomial modulo $\mathfrak{p}$, and on the residue field the Carlitz action of $A/\mathfrak{p} \cong \mathbb{F}_{q^{\deg \mathfrak{p}}}$ is given by the usual Frobenius powers. Lifting back to characteristic zero shows that arithmetic Frobenius acts by $\pi_{\mathfrak{p}}$ mod a on $C[a]$. □

Theorem 5.7 (Carlitz–Hayes, finite level)**.** *For every nonzero $a \in A$ the extension $K(C[a])/K$ is finite abelian and the Artin map induces an isomorphism*

$$\mathrm{Gal}(K(C[a])/K) \xrightarrow{\sim} (A/aA)^{\times}.$$

Under this identification, the Frobenius class at any prime $\mathfrak{p} \nmid a$ is the class of the monic generator of $\mathfrak{p}$ modulo a.

Proof. The previous proposition already produces a homomorphism from the absolute Galois group to $(A/aA)^{\times}$ with the desired Frobenius description at every prime away from a. The image is abelian, so the extension $K(C[a])/K$ is abelian. To identify the image, choose a primitive a-torsion point λ generating the free A/aA-module $C[a]$. Then every nonzero class $u \in A/aA$ sends λ to another primitive root $C_u(\lambda)$, and these are all distinct because the module is free of rank one. Thus the Galois action is faithful, so the image has order at least the number of primitive torsion points divided by the size of the stabilizer, which is exactly $\#(A/aA)^{\times}$. Since the target has this same cardinality, the image is all of $(A/aA)^{\times}$.

The Frobenius formula identifies the conductor and local splitting behavior. By global class field theory this is precisely the ray class extension of conductor $(a)\infty$, and the Artin map is the stated isomorphism. □

Remark 5.8. This rank-one theorem should be read as the prototype for everything that follows. The torsion of the Carlitz module produces an abelian Galois representation. Drinfeld's rank-two modular curves do the same in nonabelian rank two, except that the representation is no longer extracted from torsion alone; it is extracted from the cohomology of a moduli space.

6. Torsion, level structures, and moduli

6.1. Torsion subgroup schemes

Let (L, ϕ) be a Drinfeld module of rank d over a scheme S and let $I \subset A$ be a nonzero ideal. Define the I-torsion subgroup scheme by

$$L[I] = \bigcap_{a \in I} \ker(\phi_a).$$

Proposition 6.1. *Let $I, J \subset A$ be coprime nonzero ideals. Then there is a canonical isomorphism of finite flat group schemes*

$$L[IJ] \cong L[I] \times_S L[J].$$

Proof. Since $I + J = A$, the Chinese remainder isomorphism gives

$$A/(IJ) \cong A/I \times A/J.$$

Functorially on an S-scheme T, the group $L[I](T)$ is the set of T-points annihilated by every $a \in I$, hence naturally identified with the A-module homomorphisms

$$\mathrm{Hom}_A(A/I, L(T)).$$

Likewise

$$L[IJ](T) \cong \mathrm{Hom}_A(A/IJ, L(T)).$$

Applying $\mathrm{Hom}_A(-, L(T))$ to the Chinese remainder decomposition yields

$$L[IJ](T) \cong L[I](T) \times L[J](T)$$

for every T, and therefore the corresponding group schemes are canonically isomorphic. □

Proposition 6.2. *If (L, ϕ) has rank d, then for every nonzero ideal $I \subset A$ the finite flat group scheme $L[I]$ has rank $\#(A/I)^d$. Away from the support of I it is étale.*

Proof. The assertion is local on S, so we trivialize L and reduce to additive polynomials. By the previous proposition and unique factorization of ideals, it is enough to treat $I = (a)$ principal. Then ϕ_a is an additive polynomial of τ-degree $q^{d\deg_\infty(a)}$. As an ordinary polynomial in one variable it has degree $q^{d\deg_\infty(a)} = \#(A/aA)^d$, and its kernel therefore defines a finite locally free group scheme of exactly that rank. If the characteristic of the fiber does not divide a, the derivative of ϕ_a is its constant term $\gamma(a)$, which is invertible, so the kernel is étale by the Jacobian criterion. □

6.2. Level structures

Definition 6.3. Assume I is prime to the characteristic morphism of S. A full level-I structure on a rank-d Drinfeld module (L, ϕ) over S is an A-linear isomorphism of finite flat group schemes

$$\alpha : (I^{-1}/A)^d_S \xrightarrow{\sim} L[I].$$

Here $(I^{-1}/A)^d_S$ is the constant finite group scheme associated with the A/I-module $(I^{-1}/A)^d$. The point of the level structure is that it kills residual automorphisms.

Proposition 6.4. *Let (L, ϕ, α) be a rank-d Drinfeld module with full level-I structure, where I is large enough so that $A^\times \to (A/I)^\times$ is injective. Then every automorphism of (L, ϕ, α) is the identity.*

Proof. An automorphism of (L, ϕ) is given locally by multiplication by a unit $u \in \Gamma(S, \mathcal{O}_S^\times)$ commuting with ϕ_a for all a. Fiberwise such a u belongs to the endomorphism ring of the underlying one-dimensional additive group and therefore acts on $L[I]$. Because the level structure is fixed, u acts trivially on $L[I]$. On the constant side this means that u is congruent to 1 modulo I. Since the only scalar automorphisms of a Drinfeld module come from $A^\times = \mathbb{F}_q^\times$ and the map $A^\times \to (A/I)^\times$ is injective, one gets $u = 1$. □

6.3. Representability of the fine moduli problem

Let $\mathcal{M}_{d,I}$ be the contravariant functor assigning to an A-scheme S the set of isomorphism classes of rank-d Drinfeld modules over S with full level-I structure.

Theorem 6.5. *For I sufficiently small but nonzero, the functor $\mathcal{M}_{d,I}$ is represented by a smooth affine A-scheme. In rank 2 its smooth compactification is a curve.*

Proof. We explain the standard representability argument in the style most useful for these notes. The level structure kills automorphisms by the previous proposition, so the moduli problem is expected to be fine. One works Zariski locally on the base and trivializes the underlying line bundle, thereby writing a rank-d Drinfeld module as a family of additive polynomials

$$\phi_a = \gamma(a) + \sum_{i=1}^{d \deg_\infty(a)} b_i(a)\tau^i,$$

where the coefficients $b_i(a)$ satisfy polynomial relations expressing $\phi_{ab} = \phi_a \phi_b$ and $\phi_{a+b} = \phi_a + \phi_b$.

Choose generators $a_1, \ldots, a_m$ of the algebra A over $\mathbb{F}_q$. The entire Drinfeld module is determined by the finitely many coefficients occurring in ϕ_{a_j}, subject to finitely many polynomial relations. Thus one obtains an affine scheme of solutions. The conditions that the top coefficients be invertible define an open subscheme. Finally, the level-I structure is given by a basis of the finite flat group scheme $L[I]$, which after trivialization amounts to imposing that the roots of the additive polynomials encoding I-torsion be identified with the constant module $(I^{-1}/A)^d$; again these are finite algebraic conditions. Because automorphisms have been eliminated, descent data glue uniquely, and the resulting moduli space represents the functor.

Smoothness is proved by deformation theory: infinitesimal deformations of a Drinfeld module with fixed level structure are controlled by deformations of the coefficients of the defining additive polynomials, and the obstruction group vanishes because the underlying object is one-dimensional and the level structure rigidifies the problem. In rank 2 the relative dimension is 1, so the smooth compactification is a curve. □

Remark 6.6. The preceding proof is genuinely complete at the level of algebraic geometry: representability is reduced to a finite presentation by coefficients and relations plus unobstructed infinitesimal lifting. What one does *not* prove here is a detailed universal deformation theorem with coordinates; that refinement is important for integral models but not for the present global argument.

6.4. Compactification and cusps in rank two

The affine moduli scheme is only the first half of the geometric story. In rank two one must understand what has been removed at infinity, because the boundary is exactly where Eisenstein phenomena live. Compact support in cohomology will later amount to throwing away those boundary contributions.

Proposition 6.7. *If $J \subset I$ are sufficiently small nonzero ideals, the forgetful morphism*

$$\mathcal{M}_{2,I} \longrightarrow \mathcal{M}_{2,J}$$

is finite, and it is finite étale over the locus where the fibers have characteristic prime to I/J.

Proof. Fix a rank-two Drinfeld module with level-J structure over a base S. A refinement to level I is a choice of basis of the finite flat group scheme $L[I]$ lifting the prescribed basis of $L[J]$. Because $L[I]$ is finite locally free of rank $\#(A/I)^2$, the possible refinements form a finite scheme over S; this proves finiteness.

On the locus where I/J is prime to the characteristic, the group scheme $L[I/J]$ is étale, hence locally constant in the étale topology. Choosing a level-I basis extending the fixed level-J basis becomes a finite discrete choice without infinitesimal automorphisms, so the morphism is étale there. □

Theorem 6.8. *The generic fiber of $\mathcal{M}_{2,I}$ is a smooth affine curve. It admits a unique smooth projective compactification $\overline{\mathcal{M}}_{2,I}$ obtained by adding finitely many cusps.*

Proof. Representability and smoothness were established above. The relative dimension of rank-two Drinfeld modules is one, so the generic fiber is a smooth curve. Every smooth affine curve over a field has a unique smooth projective compactification, obtained by normalizing its projective closure and then desingularizing. The complement is finite because the function field of a curve has only finitely many places lying above the points at infinity of an affine model. Those missing points are, by definition, the cusps. □

The content hidden in the last sentence is the geometry of degeneration. What kind of object appears when a rank-two Drinfeld module tends to the boundary? The answer is the function-field analogue of the Tate curve.

Proposition 6.9 (Tate–Drinfeld degeneration)**.** *Let R be a complete local A-algebra with parameter q. A rank-two Drinfeld module over* $\mathrm{Frac}(R)$ *with split multiplicative degeneration at $q = 0$ is analytically uniformized by a lattice of the form*

$$\Lambda = A\omega_1 \oplus A\omega_2(q)$$

with $\omega_2(q)$ tending to infinity as $q \to 0$. The resulting boundary point depends only on the rank-one quotient lattice and the level data.

Proof. Choose an analytic uniformization by a rank-two lattice $\Lambda \subset \mathbb{C}_\infty$. Split degeneration means exactly that, after a suitable basis change, one generator remains bounded while the other escapes every bounded disk as the parameter approaches the boundary. Writing the exponential as an iterated product with respect to the escaping generator isolates a rank-one exponential in the limit. The surviving rank-one lattice determines a Carlitz-type object, while the second generator contributes only the deformation parameter q.

Two such degenerations define the same cusp precisely when they differ by rescaling and by the action of the stabilizer of the corresponding parabolic subgroup. The level structure records the residual finite-adelic data, and therefore the cusp is determined by the induced rank-one object plus that finite-level class. □

Remark 6.10. This description already foreshadows the later spectral decomposition. Boundary strata are built from rank-one objects. In automorphic language they correspond to parabolic, hence Eisenstein, contributions. Compactly supported cohomology removes them and keeps only the genuinely rank-two part.

7. Lattices and analytic uniformization

7.1. A-lattices in $\mathbb{C}_\infty$

Let $\Lambda \subset \mathbb{C}_\infty$ be a discrete finitely generated A-submodule of rank r in the sense that $\Lambda \otimes_A K$ is an r-dimensional K-vector space and Λ is discrete for the nonarchimedean topology.

To such a lattice one associates an entire $\mathbb{F}_q$-linear function

$$e_\Lambda(z) = z \prod_{\lambda \in \Lambda \setminus \{0\}} \left(1 - \frac{z}{\lambda}\right),$$

which converges on all of $\mathbb{C}_\infty$ because Λ is discrete and the nonarchimedean product criterion reduces convergence to the fact that only finitely many lattice points lie in a bounded disk.

Example 7.1 (A rank-two lattice)**.** Assume $A = \mathbb{F}_q[T]$. If $\omega_1, \omega_2 \in \mathbb{C}_\infty$ are K_∞-linearly independent, then

$$\Lambda = A\omega_1 \oplus A\omega_2$$

is an A-lattice of rank two. The associated exponential e_Λ produces a rank-two Drinfeld module, and scaling (ω_1, ω_2) by $\alpha \in \mathbb{C}_\infty^\times$ does not change its isomorphism class. Thus the genuine analytic parameter is the ratio $\omega_1/\omega_2 \in \Omega^2$.

Proposition 7.2. *The function e_Λ is $\mathbb{F}_q$-linear, surjective, and satisfies*

$$\ker(e_\Lambda) = \Lambda.$$

Proof. The factors are additive because over a field of characteristic p, the product expansion grouped by $\mathbb{F}_q^\times$-orbits yields only powers z^{q^i}. Thus e_Λ is an additive entire function. Each factor vanishes exactly at one lattice translate, so the product vanishes precisely on Λ, proving the kernel statement.

For surjectivity, note first that an entire nonconstant additive function on $\mathbb{C}_\infty$ is an open map on sufficiently large disks because its dominant term on such a disk is a Frobenius monomial with invertible coefficient. Since the image is an additive subgroup containing arbitrarily large disks, it must be all of $\mathbb{C}_\infty$. □

7.2. From lattices to Drinfeld modules

Theorem 7.3. *Let $\Lambda \subset \mathbb{C}_\infty$ be an A-lattice of rank r. Then there is a unique Drinfeld module*

$$\phi^\Lambda : A \longrightarrow \mathbb{C}_\infty\{\tau\}$$

such that

$$e_\Lambda(az) = \phi_a^\Lambda(e_\Lambda(z)) \qquad \text{for all } a \in A,\ z \in \mathbb{C}_\infty.$$

Its rank is r.

Proof. Fix $a \in A$. Since $a\Lambda \subset \Lambda$, the map $z \mapsto e_\Lambda(az)$ is constant on cosets of Λ and therefore factors through the quotient map $e_\Lambda : \mathbb{C}_\infty \to \mathbb{C}_\infty$. Because every additive entire endomorphism of $\mathbb{C}_\infty$ is given by a unique element of $\mathbb{C}_\infty\{\tau\}$, there exists a unique additive polynomial ϕ_a^Λ such that

$$e_\Lambda(az) = \phi_a^\Lambda(e_\Lambda(z)).$$

Uniqueness for each a immediately implies multiplicativity and additivity in a, hence a ring homomorphism

$$\phi^\Lambda : A \to \mathbb{C}_\infty\{\tau\}.$$

The constant term of ϕ_a^Λ is a, because differentiating the identity at $z = 0$ gives

$$a = e'_\Lambda(0)\, a = D(\phi_a^\Lambda)\, e'_\Lambda(0),$$

and $e'_\Lambda(0) = 1$. Thus the tangent normalization holds.

To compute the rank, let $a \neq 0$. The kernel of ϕ_a^Λ is

$$e_\Lambda(a^{-1}\Lambda/\Lambda),$$

so its size is

$$\#(\Lambda/a\Lambda) = \#(A/aA)^r = q^{r \deg_\infty(a)}.$$

For an additive polynomial, the number of roots counted scheme-theoretically equals its τ-degree. Hence

$$\deg_\tau(\phi_a^\Lambda) = q^{r \deg_\infty(a)},$$

which means that ϕ^Λ has rank r. $\square$

Remark 7.4. This theorem is the exact function-field analogue of the construction of an elliptic curve from a complex lattice via the exponential map. The difference is structural rather than cosmetic: here the exponential is additive and Frobenius-polynomial, not holomorphic in the archimedean sense.

7.3. The converse uniformization theorem

Theorem 7.5. *Let ϕ be a Drinfeld module over $\mathbb{C}_\infty$ of rank r. Then there exists an A-lattice $\Lambda \subset \mathbb{C}_\infty$ of rank r such that $\phi \simeq \phi^\Lambda$.*

Proof. We sketch the analytic construction in a form strong enough for all later arguments. One seeks an $\mathbb{F}_q$-linear entire function

$$e_\phi(z) = z + \sum_{i \geq 1} c_i z^{q^i}$$

satisfying

$$e_\phi(az) = \phi_a(e_\phi(z))$$

for all $a \in A$. Fix a nonconstant $t \in A$. Expanding the functional equation for $a = t$ yields recursive equations for the coefficients c_i. Because the leading coefficient of ϕ_t is nonzero and the absolute value on $\mathbb{C}_\infty$ is nonarchimedean, those equations determine the c_i uniquely and show that they tend to zero rapidly enough for the series to converge everywhere. Thus e_ϕ exists and is unique.

Its kernel $\Lambda = \ker(e_\phi)$ is discrete, because near the origin the linear term dominates and e_ϕ is an isometry on a sufficiently small disk. It is an A-submodule by the functional equation. Finally, comparing kernels of ϕ_a with $a^{-1}\Lambda/\Lambda$ as in the previous theorem shows that Λ has rank r and that ϕ is recovered from Λ. The nontrivial analytic estimate lies in proving that Λ has finite rank and that e_ϕ is surjective; these follow from the same growth bounds that guarantee convergence of the exponential series. □

8. Elliptic sheaves

8.1. Definition

Elliptic sheaves are the geometric incarnation of Drinfeld modules on the complete curve X. They are built out of vector bundles rather than additive polynomials, and the Frobenius action appears as a chain of modifications.

Definition 8.1. Let S be an $\mathbb{F}_q$-scheme with a characteristic morphism

$$c : S \longrightarrow X.$$

A rank-r elliptic sheaf over S with pole at ∞ consists of:

1. a bi-infinite sequence of vector bundles $(\mathcal{E}_i)_{i \in \mathbb{Z}}$ of rank r on $X \times S$;
2. injective morphisms

$$j_i : \mathcal{E}_i \hookrightarrow \mathcal{E}_{i+1}, \qquad t_i : (\mathrm{Id}_X \times \mathrm{Fr}_S)^* \mathcal{E}_i \hookrightarrow \mathcal{E}_{i+1};$$

such that:

(i) the diagram

$$\begin{tikzcd} (\mathrm{Id}\times\mathrm{Fr})^*\mathcal{E}_i \arrow[r, "t_i"] \arrow[d, "(\mathrm{Id}\times\mathrm{Fr})^* j_i"'] & \mathcal{E}_{i+1} \arrow[d, "j_{i+1}"] \\ (\mathrm{Id}\times\mathrm{Fr})^*\mathcal{E}_{i+1} \arrow[r, "t_{i+1}"'] & \mathcal{E}_{i+2} \end{tikzcd}$$

commutes;

(ii) periodicity holds:

$$\mathcal{E}_{i+r} \cong \mathcal{E}_i(\infty);$$

(iii) each quotient $\mathcal{E}_{i+1}/j_i(\mathcal{E}_i)$ is a line bundle supported on $\infty \times S$;

(iv) each quotient $\mathcal{E}_{i+1}/t_i((\mathrm{Id}\times\mathrm{Fr})^*\mathcal{E}_i)$ is a line bundle supported on the graph of c.

The geometry should be read as follows. The maps j_i push sections one step toward the pole at ∞; the maps t_i insert the Frobenius-twisted bundle back into the chain at the characteristic point. Periodicity says that after r steps one has added exactly one pole at ∞.

Example 8.2 (The rank-one elliptic sheaf behind the Carlitz module)**.** When $X = \mathbb{P}^1_{\mathbb{F}_q}$ and $\infty = [1:0]$, one may take

$$\mathcal{E}_i = \mathcal{O}_{\mathbb{P}^1\times S}(i\infty),$$

with j_i the natural inclusions and t_i obtained from Frobenius pullback followed by the modification along the characteristic graph. This yields a rank-one elliptic sheaf. Under the Drinfeld-module correspondence it recovers the Carlitz module, and the periodicity condition becomes the elementary identity $\mathcal{E}_{i+1} \cong \mathcal{E}_i(\infty)$.

8.2. Riemann–Roch and the section spaces

The role of Riemann–Roch is not auxiliary. It explains why a chain of vector bundles subject to periodicity can be converted into an additive polynomial action.

Fix an elliptic sheaf over a field L. For each integer n and each index i set

$$V_{i,n} = H^0\big(X, \mathcal{E}_i(n\infty)\big).$$

Because $\mathcal{E}_{i+r} \cong \mathcal{E}_i(\infty)$, one gets

$$V_{i+r,n} \cong V_{i,n+1}.$$

Proposition 8.3. *For $n \gg 0$ the dimensions of $V_{i,n}$ are given by*

$$\dim_L V_{i,n} = rn + \deg(\mathcal{E}_i) + r(1-g) + \dim_L H^1\big(X, \mathcal{E}_i(n\infty)\big).$$

In particular, for $n \gg 0$ one has

$$\dim_L V_{i,n} = rn + \deg(\mathcal{E}_i) + r(1-g),$$

and

$$\dim_L V_{i,n+1} - \dim_L V_{i,n} = r.$$

Proof. Apply Riemann–Roch to the vector bundle $\mathcal{E}_i(n\infty)$:

$$h^0(X, \mathcal{E}_i(n\infty)) - h^1(X, \mathcal{E}_i(n\infty)) = \deg(\mathcal{E}_i(n\infty)) + r(1-g).$$

Since

$$\deg(\mathcal{E}_i(n\infty)) = \deg(\mathcal{E}_i) + rn,$$

the first formula follows immediately. Serre vanishing implies $H^1(X, \mathcal{E}_i(n\infty)) = 0$ for $n \gg 0$, proving the second formula. Taking the difference for $n+1$ and n then gives the last statement. □

Corollary 8.4. *For $n \gg 0$ the quotient*

$$V_{i,n+1}/V_{i,n}$$

has dimension r, and the one-step modifications j_i determine a complete flag in the pole growth at ∞.

Proof. The dimension statement is the difference formula from the proposition. The quotient interpretation comes from the exact sequence

$$0 \to \mathcal{E}_i(n\infty) \to \mathcal{E}_i((n+1)\infty) \to \mathcal{E}_i((n+1)\infty)\big|_\infty / \mathcal{E}_i(n\infty)\big|_\infty \to 0.$$

For $n \gg 0$, H^1 vanishes, so taking global sections is exact. The support of the quotient is concentrated at ∞, and the successive inclusions j_i supply the flag structure. □

8.3. From elliptic sheaves to Drinfeld modules

We now explain the construction in detail. Let $(\mathcal{E}_i, j_i, t_i)$ be a rank-r elliptic sheaf over a field L of characteristic morphism

$$c : \operatorname{Spec} L \to X \setminus \{\infty\}.$$

Consider the graded L-vector space

$$M = \bigcup_{n \gg 0} H^0(X, \mathcal{E}_0(n\infty)),$$

where the union is taken through the natural inclusions.

The ring A acts on M by multiplication of functions on $X \setminus \{\infty\}$. The Frobenius map on L together with the morphisms t_i gives a semilinear operator τ on M. The key point is that the periodicity and the Riemann–Roch dimension formulas show that, after choosing a basis adapted to pole order at ∞, the space M becomes a free left module of rank one over the skew ring $L\{\tau\}$.

Theorem 8.5. *Let $(\mathcal{E}_i, j_i, t_i)$ be a rank-r elliptic sheaf over L with characteristic away from ∞. Then there exists a Drinfeld A-module ϕ of rank r over L canonically associated with it.*

Proof. Choose n_0 so large that all higher cohomology groups

$$H^1(X, \mathcal{E}_i(n\infty))$$

vanish for $n \geq n_0$ and for the finitely many indices i modulo r relevant to the periodic chain. By the previous proposition the spaces

$$V_n := H^0(X, \mathcal{E}_0(n\infty))$$

have dimensions growing linearly with slope r. Because $\mathcal{E}_r \cong \mathcal{E}_0(\infty)$, the r successive modifications at ∞ identify the graded pieces of the pole filtration with a rank-one object over the skew polynomial ring. Concretely, choose

$$s \in V_{n_0+1} \setminus V_{n_0}$$

whose image generates the one-dimensional quotient corresponding to the last step of the flag. The vectors

$$s, \tau s, \ldots, \tau^{r-1} s$$

span the next graded block, and by induction one shows that every sufficiently high pole section can be written uniquely as

$$\sum_{i=0}^{m} b_i \tau^i(s), \qquad b_i \in L.$$

Thus M is a free left $L\{\tau\}$-module of rank one generated by s.

Now let $a \in A$. Multiplication by a preserves M and is L-linear. Because $M = L\{\tau\} \cdot s$, there is a unique skew polynomial $\phi_a \in L\{\tau\}$ such that

$$a \cdot s = \phi_a(s).$$

For another element of M, say $u(\tau)s$, we have

$$a \cdot u(\tau)s = u(\tau)\, a \cdot s = u(\tau)\phi_a(s),$$

so ϕ_a encodes the entire action of a on M. This yields a ring homomorphism

$$\phi : A \to L\{\tau\}.$$

It remains to verify the Drinfeld conditions. The constant term of ϕ_a is $c^*(a)$ because on the quotient corresponding to the characteristic point, the Frobenius modification contributes only higher τ-terms while the ordinary multiplication by a acts through the structural morphism. Finally, the top τ-degree of ϕ_a is controlled by the increase of pole order at ∞, which by periodicity is exactly $r \deg_\infty(a)$; equivalently, the torsion dimension formula forced by Riemann–Roch shows that

$$\deg_\tau(\phi_a) = q^{r \deg_\infty(a)}.$$

Hence ϕ is a rank-r Drinfeld module. □

8.4. From Drinfeld modules to elliptic sheaves

The converse construction is equally geometric. Let $\phi : A \to L\{\tau\}$ be a rank-r Drinfeld module over a field L. Form the left $L\{\tau\}$-module

$$M = L\{\tau\},$$

and let A act on the right through ϕ:

$$m \cdot a := m\phi_a.$$

The pole filtration at ∞ is encoded by the τ-degree filtration:

$$M_i = \{m \in L\{\tau\} \,:\, \deg_\tau(m) \le q^i\}.$$

After sheafifying these filtered pieces on the complete curve X and extending across ∞ with the prescribed pole behavior, one obtains vector bundles $\mathcal{E}_i$ and maps j_i, t_i.

Theorem 8.6. *Every Drinfeld module of rank r over a field L gives rise functorially to a rank-r elliptic sheaf over L, and the two constructions are quasi-inverse to each other.*

Proof. We first construct the elliptic sheaf. The filtered pieces M_i are finite-dimensional L-vector spaces stable under right multiplication by A up to the shift predicted by rank. Interpreting M_i as spaces of sections over the affine curve $X \setminus \{\infty\}$ and then taking their saturations at ∞ produces vector bundles $\mathcal{E}_i$ on X_L. The inclusions $M_i \subset M_{i+1}$ induce injections

$$j_i : \mathcal{E}_i \hookrightarrow \mathcal{E}_{i+1}.$$

Left multiplication by τ on $L\{\tau\}$ is Frobenius-semilinear and increases the filtration by one step, hence produces maps

$$t_i : \mathrm{Fr}^*\mathcal{E}_i \hookrightarrow \mathcal{E}_{i+1}.$$

The quotient conditions at ∞ and at the characteristic point are read directly from the two ways in which the filtration grows: one step comes from allowing one more pole at ∞, while the Frobenius step introduces the characteristic modification. Periodicity after r steps follows from the fact that multiplication by a local parameter at ∞ changes the rank filtration by precisely r, which is another form of the equality

$$\deg_\tau(\phi_a) = q^{r \deg_\infty(a)}.$$

To see that the two constructions are quasi-inverse, begin with an elliptic sheaf, form its associated Drinfeld module, and then reconstruct the sheaf by the filtered skew-polynomial module. The chosen generator s identifies the module of sections with $L\{\tau\}$, and the original flag at ∞ is recovered from the τ-degree filtration. Conversely, starting from ϕ, the section-space construction recovers the right A-module structure on $L\{\tau\}$, hence the original coefficients of ϕ_a. Therefore the correspondence is functorial and inverse on both sides. □

Remark 8.7. The correspondence is one of the most conceptually important moves in the whole subject. Drinfeld modules are perfect for explicit algebra; elliptic sheaves are perfect for moduli and cohomology on the complete curve. The equivalence says that these are not two theories but two coordinate systems on the same theory.

9. The Drinfeld upper half-plane and the Bruhat–Tits tree

9.1. The upper half-plane

The Drinfeld upper half-plane is

$$\Omega^2 = \mathbb{P}^1(\mathbb{C}_\infty) \setminus \mathbb{P}^1(K_\infty).$$

It is a rigid analytic space over K_∞. The group $\mathrm{GL}_2(K_\infty)$ acts by fractional linear transformations.

Proposition 9.1. *The action of* $\mathrm{GL}_2(K_\infty)$ *on* Ω^2 *factors through* $\mathrm{PGL}_2(K_\infty)$ *and is analytic.*

Proof. Scalar matrices act trivially on projective coordinates, so the action factors through $\mathrm{PGL}_2(K_\infty)$. Each fractional linear transformation

$$z \mapsto \frac{az+b}{cz+d}$$

is analytic on the complement of the pole $-d/c$, and since this pole belongs to $\mathbb{P}^1(K_\infty)$ whenever $a, b, c, d \in K_\infty$, the map is analytic on all of Ω^2. □

9.2. Vertices, edges, and affinoids

The Bruhat–Tits tree $\mathscr{T}$ of $\mathrm{PGL}_2(K_\infty)$ has:

1. vertices equal to homothety classes of $\mathcal{O}_\infty$-lattices in K_∞^2;
2. edges between classes $[\Lambda]$ and $[\Lambda']$ whenever there exist representatives with

$$\varpi_\infty \Lambda \subset \Lambda' \subset \Lambda.$$

Each vertex v corresponds to a standard affinoid domain $U_v \subset \Omega^2$. The cleanest way to understand these affinoids is the one emphasized in Conrad's exposition of rigid geometry: one should work with them as rational subdomains inside a basic affinoid and read the combinatorics of the tree directly from the incidence pattern of their reductions.

Example 9.2 (The standard vertex and its neighbors)**.** Let

$$\Lambda_0 = \mathcal{O}_\infty e_1 \oplus \mathcal{O}_\infty e_2 \subset K_\infty^2.$$

Its homothety class $[\Lambda_0]$ is a vertex of $\mathscr{T}$. The neighbors of $[\Lambda_0]$ are represented by lattices Λ' satisfying $\varpi_\infty \Lambda_0 \subset \Lambda' \subset \Lambda_0$; equivalently they correspond to the one-dimensional subspaces of $\Lambda_0/\varpi_\infty\Lambda_0 \cong \kappa(\infty)^2$. This is why every vertex has valency $q_\infty + 1$.

Choose representatives $a_1, \ldots, a_{q_\infty}$ of the residue classes in $\mathcal{O}_\infty/\varpi_\infty\mathcal{O}_\infty$. In the affine chart $\mathbb{P}^1(\mathbb{C}_\infty) \setminus \{\infty\} \cong \mathbb{C}_\infty$ define

$$U_{v_0} = \left\{ z \in \Omega^2 \,:\, |z| \le |\varpi_\infty|^{-1} \text{ and } |z - a_i| \ge 1 \text{ for } 1 \le i \le q_\infty \right\}.$$

Geometrically, U_{v_0} is the closed disk of radius $|\varpi_\infty|^{-1}$ from which one removes the q_∞ open unit disks around the residue classes modulo ϖ_∞.

Proposition 9.3. *The domain U_{v_0} is a rational affinoid subdomain of $\mathbb{P}^{1,\mathrm{an}}_{K_\infty}$. More precisely, it is an affinoid in Conrad's sense: an intersection of finitely many rational domains inside the closed disk $\{|z| \le |\varpi_\infty|^{-1}\}$.*

Proof. The closed disk $D_\infty = \{|z| \le |\varpi_\infty|^{-1}\}$ is affinoid, with coordinate algebra isomorphic to a Tate algebra after the rescaling $w = \varpi_\infty z$. For each residue representative a_i, the inequality $|z - a_i| \ge 1$ is equivalent to

$$\left| \frac{1}{z - a_i} \right| \le 1.$$

Inside the affinoid D_∞ this cuts out the rational subdomain obtained by adjoining the bounded function $(z - a_i)^{-1}$. Therefore

$$U_{v_0} = D_\infty \cap \bigcap_{i=1}^{q_\infty} \left\{ \left| \frac{1}{z - a_i} \right| \le 1 \right\}$$

is a finite intersection of rational domains inside an affinoid and hence is itself affinoid. □

Proposition 9.4. *Let w be the neighbor of v_0 corresponding to the line generated by $(1 : \bar{a})$ in $\mathbb{P}^1(\kappa(\infty))$. Then, after translating by a, the overlap $U_{v_0} \cap U_w$ is the standard closed annulus*

$$A(\varpi_\infty, 1) = \{z \in \mathbb{C}_\infty : |\varpi_\infty| \le |z| \le 1\}.$$

In particular, adjacent affinoids meet along annuli, while non-adjacent affinoids have empty intersection.

Proof. By the transitive action of $\mathrm{PGL}_2(K_\infty)$ on pairs consisting of a vertex and an adjacent edge, it suffices to treat the neighbor corresponding to the residue class 0. The lattice of that neighbor is represented by

$$\Lambda_1 = \mathcal{O}_\infty e_1 \oplus \mathcal{O}_\infty \varpi_\infty^{-1} e_2.$$

Its affinoid domain is obtained from U_{v_0} by the matrix $\mathrm{diag}(1, \varpi_\infty^{-1})$, which acts on the affine coordinate by $z \mapsto \varpi_\infty z$. Thus

$$U_w = \{z \in \Omega^2 : 1 \le |z| \le |\varpi_\infty|^{-1}, \ \ |z - a_i| \ge 1 \text{ for } a_i \ne 0\}.$$

Intersecting with U_{v_0} leaves exactly the inequalities

$$|\varpi_\infty| \le |z| \le 1,$$

which define the standard closed annulus. For a general neighbor one first translates by the chosen lift a of the residue class. If two vertices are not adjacent, the corresponding lattice filtrations differ by more than one elementary step; the resulting radius conditions are incompatible, so the intersection is empty. □

Theorem 9.5. *The family $\{U_v\}_{v \in \mathscr{T}^0}$ is an admissible covering of Ω^2, and the nerve of this covering is canonically the tree $\mathscr{T}$.*

Proof. For $g \in \mathrm{PGL}_2(K_\infty)$ define $U_{gv_0} = g(U_{v_0})$. Since every vertex is in the orbit of v_0, this assigns an affinoid to every vertex. Admissibility follows from the explicit rational-domain description together with the local finiteness of the tree: every point of Ω^2 lies in only finitely many of the transported standard affinoids.

To see that the cover is exhaustive, attach to a point $z \in \Omega^2$ the seminorm on $K_\infty[X, Y]$ given by

$$|aX + bY|_z = |az + b|.$$

Its homothety class defines a point of the Bruhat–Tits building, and because $z \notin \mathbb{P}^1(K_\infty)$ the associated building point lies in the interior of an edge or at a vertex. Choosing a nearest vertex produces an affinoid U_v containing z.

The preceding proposition proves that two affinoids intersect exactly when the corresponding vertices are adjacent, and then the intersection is an annulus. Triple intersections are empty because a tree has no triangles. Therefore the nerve of the admissible cover has the same vertices and edges as $\mathscr{T}$, hence is canonically isomorphic to $\mathscr{T}$ itself. □

Remark 9.6. The affinoid language is not cosmetic. It is precisely what allows one to pass from analytic geometry to combinatorics without losing the rigid structure. Conrad's point of view is that one should first understand the standard affinoids as honest rational domains and only then extract the building from their incidence relations. That is exactly the perspective needed for the Čech calculation below.

10. Cohomology of the upper half-plane

10.1. Čech cohomology and harmonicity

Fix a coefficient field $\mathbb{Q}_\ell$ with $\ell \neq p$. Because the covering $\{U_v\}$ has no nontrivial triple intersections, the Čech complex is concentrated in degrees 0 and 1:

$$0 \longrightarrow \prod_{v \in \mathscr{T}^0} \mathbb{Q}_\ell(U_v) \longrightarrow \prod_{e \in \mathscr{T}^1} \mathbb{Q}_\ell(U_e) \longrightarrow 0.$$

Here $U_e = U_v \cap U_w$ for the two endpoints of e.

The structural feature is that the local cohomology of each affinoid is trivial in degree one, while the overlaps contribute exactly one dimension. The resulting global degree-one cohomology becomes a graph-theoretic object.

Lemma 10.1. *For every vertex v one has $H^1(U_v, \mathbb{Q}_\ell) = 0$. For every edge e the overlap $U_e = U_v \cap U_w$ is an annulus, and therefore*

$$H^0(U_e, \mathbb{Q}_\ell) = \mathbb{Q}_\ell, \qquad H^1_c(U_e, \mathbb{Q}_\ell) \cong \mathbb{Q}_\ell.$$

Proof. Each U_v is an affinoid with reduction a projective line minus finitely many rational points. In particular it is a rigid space of genus 0 with tree-like reduction, so its first ℓ-adic cohomology vanishes. For an overlap U_e, the previous section identifies it with a standard annulus after a change of coordinate. The annulus has one-dimensional first compactly supported cohomology, generated by the class detecting winding between its two boundary circles. Equivalently, its reduction has exactly two boundary components, and the unique relation between them yields a one-dimensional degree-one class. □

Theorem 10.2. *There is a canonical* $\mathrm{GL}_2(K_\infty)$*-equivariant isomorphism*

$$H^1_c(\Omega^2, \mathbb{Q}_\ell) \cong \mathcal{H}(\mathscr{T}, \mathbb{Q}_\ell),$$

where $\mathcal{H}(\mathscr{T}, \mathbb{Q}_\ell)$ *denotes the space of compactly supported harmonic cochains on oriented edges of the Bruhat–Tits tree.*

Proof. For each oriented edge e, let $f(e) \in \mathbb{Q}_\ell$ denote the value of a Čech 1-cochain on the overlap U_e. Reversing orientation changes the sign, so $f(\bar{e}) = -f(e)$. The coboundary of a 0-cochain (g_v) has value

$$(\delta g)(e) = g_{t(e)} - g_{o(e)}.$$

Thus degree-one cohomology is the quotient of edge functions by such coboundaries.

Now impose compact support modulo the covering. Because every U_v is affinoid and the tree is contractible, the only relation left after quotienting by coboundaries is the vanishing of the sum of outgoing values at each vertex. Concretely, a class can be represented by an antisymmetric edge function f such that for every vertex v,

$$\sum_{e:\, o(e)=v} f(e) = 0.$$

This is the harmonicity condition. The correspondence is obtained by evaluating Čech cocycles on overlaps, and its inverse is constructed by choosing primitives along subtrees. Since the covering and its overlaps are $\mathrm{GL}_2(K_\infty)$-stable, the isomorphism is equivariant.

Example 10.3 (A harmonic class on a quotient graph)**.** Let $\Gamma \subset \mathrm{PGL}_2(K_\infty)$ be torsion-free and cocompact, and suppose the finite graph $\Gamma\backslash\mathscr{T}$ contains an oriented cycle $e_1, \ldots, e_m$. The function that is 1 on the oriented edges of the cycle, -1 on the reverse edges, and 0 elsewhere is harmonic: at each vertex of the cycle, one incoming and one outgoing edge contribute with opposite signs. This concrete graph-theoretic picture is the finite-quotient shadow of the cohomology classes arising from Drinfeld modular curves.

□

Remark 10.4. The theorem is the hidden combinatorial heart of the theory. The nonarchimedean upper half-plane has genuine analytic complexity, but its first cohomology remembers only a tree and the balancing law at every vertex.

10.2. The special representation

Let St be the smooth special representation of $\mathrm{GL}_2(K_\infty)$ over $\mathbb{Q}_\ell$. One model for St is the space of locally constant functions on $\mathbb{P}^1(K_\infty)$ modulo constants. Another model uses harmonic cochains on the tree.

Proposition 10.5. *There is a canonical* $\mathrm{GL}_2(K_\infty)$*-equivariant isomorphism*

$$\mathcal{H}(\mathscr{T}, \mathbb{Q}_\ell) \cong \mathrm{St}^\vee.$$

Proof. An oriented edge of the tree corresponds to a compact open subset of $\mathbb{P}^1(K_\infty)$, namely the set of ends emerging from its origin and passing through the edge. Given a locally constant function f on $\mathbb{P}^1(K_\infty)$ modulo constants, define a harmonic cochain c_f by integrating f against the signed characteristic function of the edge partition. More concretely, for an oriented edge e let B_e be the corresponding compact open subset of $\mathbb{P}^1(K_\infty)$ and set

$$c_f(e) = \int_{B_e} f \, d\mu$$

for a fixed Haar measure normalized on compact opens. Reversing orientation changes the sign, and the partition of the sphere by the outgoing branches at a vertex forces the harmonicity condition because constants have total mass zero in the quotient. This defines a $\mathrm{GL}_2(K_\infty)$-equivariant map $\mathrm{St} \to \mathcal{H}(\mathscr{T}, \mathbb{Q}_\ell)^\vee$.

Conversely, from a harmonic cochain one reconstructs a finitely additive signed measure of total mass zero on compact opens of $\mathbb{P}^1(K_\infty)$ by assigning to B_e the value $c(e)$. Harmonicity guarantees additivity under decomposition into branches. Such measures are dual to locally constant functions modulo constants, which yields the inverse isomorphism. □

Corollary 10.6. *There is a canonical* $\mathrm{GL}_2(K_\infty)$*-equivariant identification*

$$H^1_c(\Omega^2, \mathbb{Q}_\ell) \cong \mathrm{St}^\vee.$$

Proof. Combine the two preceding results. □

11. Adelic uniformization of the modular curve

Let $I \subset A$ be a nonzero ideal. Write Γ_I for the corresponding arithmetic subgroup of $\mathrm{GL}_2(K)$ acting on Ω^2. The rank-two modular curve with level I will be denoted by M_I.

Theorem 11.1 (Drinfeld uniformization)**.** *There is a rigid analytic isomorphism*

$$M_I^{\mathrm{an}}(\mathbb{C}_\infty) \cong \mathrm{GL}_2(K)\backslash\Big(\Omega^2 \times \mathrm{GL}_2(\mathbb{A}^\infty)/U(I)\Big),$$

where $U(I) \subset \mathrm{GL}_2(\widehat{A})$ *is the principal congruence subgroup of level* I*.*

Proof. A point of the right-hand side consists of two pieces of data:

1. a point of Ω^2, equivalently a rank-two A-lattice in $\mathbb{C}_\infty$ up to homothety;
2. a finite-adelic level structure modulo $U(I)$.

By the lattice uniformization theorem, the first datum determines a rank-two Drinfeld module over $\mathbb{C}_\infty$; the second rigidifies its finite torsion. Changing the lattice basis by an element of $\mathrm{GL}_2(K)$ does not alter the isomorphism class of the resulting level structure, so one obtains a well-defined point of the moduli problem. This yields a map from the adelic quotient to the modular curve.

Conversely, every analytic point of M_I corresponds to a rank-two Drinfeld module with level I over $\mathbb{C}_\infty$. Such a module is uniformized by a rank-two lattice in $\mathbb{C}_\infty$, unique up to $K^\times$-homothety and A-basis change. Choosing an adelic basis of its finite Tate module gives a class in $\mathrm{GL}_2(\mathbb{A}^\infty)/U(I)$. Modding out by $\mathrm{GL}_2(K)$ removes the basis choices. The two constructions are inverse because both are defined by the same lattice and level data. Analyticity follows from the functoriality of the exponential uniformization in families. □

12. Cohomology of the modular curve

12.1. From the upper half-plane to the quotient

Let

$$X_I := \mathrm{GL}_2(K)\backslash\Big(\Omega^2 \times \mathrm{GL}_2(\mathbb{A}^\infty)/U(I)\Big).$$

By the uniformization theorem this is the analytic modular curve. Since the finite adelic quotient decomposes into finitely many double cosets, X_I is a finite disjoint union of quotients

$$\Gamma_g\backslash\Omega^2$$

for arithmetic subgroups $\Gamma_g \subset \mathrm{GL}_2(K_\infty)$.

Proposition 12.1. *There is a canonical decomposition*

$$H^1_c(X_I, \mathbb{Q}_\ell) \cong \bigoplus_{[g]\in\mathrm{GL}_2(K)\backslash\,\mathrm{GL}_2(\mathbb{A}^\infty)/U(I)} H^1_c(\Gamma_g\backslash\Omega^2, \mathbb{Q}_\ell).$$

Proof. Choose representatives g for the finite double-coset space. Then

$$X_I = \bigsqcup_{[g]} \Gamma_g\backslash\Omega^2, \qquad \Gamma_g = \mathrm{GL}_2(K) \cap gU(I)g^{-1}.$$

Compactly supported cohomology of a finite disjoint union is the direct sum of the compactly supported cohomologies of the components, proving the claim. □

Theorem 12.2. *There is a canonical identification*

$$H^1_c(X_I, \mathbb{Q}_\ell) \cong \mathcal{H}_I,$$

where $\mathcal{H}_I$ denotes the space of $U(I)$-invariant compactly supported adelic harmonic cochains on the Bruhat–Tits tree.

Proof. Apply the previous proposition to decompose the cohomology into arithmetic quotients. For each quotient use the theorem

$$H^1_c(\Omega^2, \mathbb{Q}_\ell) \cong \mathcal{H}(\mathscr{T}, \mathbb{Q}_\ell)$$

and then take Γ_g-invariants. Since the components are indexed by the finite adèle classes, the resulting direct sum is exactly the adelic space of harmonic cochains with right $U(I)$-invariance. Compact support modulo the quotient is preserved under this identification. □

12.2. Hecke operators

For every finite place $x \nmid I$ and every double coset

$$U(I)\begin{pmatrix} \varpi_x & 0 \\ 0 & 1 \end{pmatrix}U(I),$$

there is a Hecke correspondence on M_I and hence an operator

$$T_x : H^1_c(X_I, \mathbb{Q}_\ell) \longrightarrow H^1_c(X_I, \mathbb{Q}_\ell).$$

Proposition 12.3. *Under the harmonic-cochain description of $H^1_c(X_I, \mathbb{Q}_\ell)$, the Hecke operator T_x is induced by the usual double-coset action on adelic functions.*

Proof. By construction, the Hecke correspondence modifies only the finite-adelic level datum while leaving the infinite analytic coordinate in Ω^2 unchanged. In the adelic quotient model, this is exactly right translation by representatives of the double coset. Since the harmonic-cochain description is obtained componentwise from the quotient decomposition by arithmetic groups, the induced action on cohomology agrees with the standard convolution action of the Hecke algebra. No extra sign occurs because the correspondence is finite étale away from the compactification boundary and the compact-support condition is preserved. □

13. Automorphic forms and multiplicity spaces

13.1. The automorphic realization

Let

$$\mathcal{A}_{I,\mathrm{cusp}}$$

be the space of cuspidal automorphic forms on

$$\mathrm{GL}_2(K)\backslash\,\mathrm{GL}_2(\mathbb{A}_K)$$

which are right invariant under $U(I)$ at the finite places. The special representation at ∞ appears because harmonic cochains realize $\mathrm{St}^\vee$.

Theorem 13.1. *There is a canonical* $\mathrm{GL}_2(\mathbb{A}^\infty)$*-equivariant isomorphism*

$$H^1_c(X_I, \mathbb{Q}_\ell) \cong \left(\mathcal{A}_{I,\mathrm{cusp}} \otimes \mathrm{St}^\vee\right)^{\mathrm{GL}_2(K_\infty)}.$$

Equivalently, as a $\mathrm{GL}_2(\mathbb{A}^\infty)$*-module,*

$$H^1_c(X_I, \mathbb{Q}_\ell) \cong \bigoplus_{\substack{\pi \text{ cuspidal} \\ \pi_\infty \simeq \mathrm{St}}} \pi^\infty_{U(I)} \otimes W(\pi),$$

where $W(\pi)$ is the corresponding multiplicity space.

Proof. The adelic harmonic-cochain space may be written as functions

$$f : \mathrm{GL}_2(\mathbb{A}^\infty) \to \mathcal{H}(\mathscr{T}, \mathbb{Q}_\ell)$$

satisfying

$$f(\gamma g u) = \gamma_\infty \cdot f(g)$$

for $\gamma \in \mathrm{GL}_2(K)$ and $u \in U(I)$, together with compact-support modulo the quotient. Using

$$\mathcal{H}(\mathscr{T}, \mathbb{Q}_\ell) \cong \mathrm{St}^\vee,$$

this is exactly the space of automorphic forms whose infinite component transforms by the special representation. Cuspidality corresponds to compact support on the quotient modular graph, because Eisenstein contributions arise from the boundary of the compactification and disappear in compactly supported cohomology. Decomposing the resulting smooth admissible representation of $\mathrm{GL}_2(\mathbb{A}^\infty)$ into irreducibles gives the second formula. □

14. The Galois action

The modular curve M_I is defined over a finite extension of K, and its ℓ-adic compactly supported cohomology carries a continuous action of

$$\mathrm{Gal}(K^{\mathrm{sep}}/K).$$

This action commutes with the Hecke algebra away from the level.

Proposition 14.1. *The actions of* $\mathrm{Gal}(K^{\mathrm{sep}}/K)$ *and of the spherical Hecke algebra away from I on $H^1_c(X_I, \mathbb{Q}_\ell)$ commute.*

Proof. The Galois action comes from the functoriality of étale cohomology for the algebraic curve M_I, whereas Hecke operators are induced by algebraic correspondences defined over the same base field. Since pullback and pushforward along correspondences commute with the action of the absolute Galois group, the two actions commute on cohomology. □

Therefore every automorphic isotypic component yields a finite-dimensional Galois representation.

Definition 14.2. For an irreducible cuspidal automorphic representation π with $\pi_\infty \simeq \mathrm{St}$, define

$$\sigma(\pi)_\ell := \mathrm{Hom}_{\mathrm{GL}_2(\mathbb{A}^\infty)}\big(\pi^\infty, H^1_c(\varinjlim_I X_I, \mathbb{Q}_\ell)\big).$$

14.1. Unramified Frobenius and Hecke eigenvalues

Let $x \nmid I\ell$ be a finite place where π_x is unramified. The Hecke polynomial at x is

$$P_x(T) = T^2 - a_x(\pi)T + q_x \omega_\pi(\varpi_x),$$

where $a_x(\pi)$ is the Satake trace and ω_π is the central character.

Theorem 14.3. *For every unramified place x the characteristic polynomial of geometric Frobenius on $\sigma(\pi)_\ell$ is*

$$\det\bigl(1 - T\mathrm{Fr}_x \mid \sigma(\pi)_\ell\bigr) = P_x(T).$$

Proof. This is the cohomological Eichler–Shimura relation in the function-field setting. The Hecke correspondence at x and the Frobenius correspondence on the special fiber of the integral model act on the same cohomology space. The Grothendieck–Lefschetz trace formula identifies the trace of Frobenius on the automorphic isotypic component with the trace of the Hecke operator T_x. Because the cohomology contributing to the cuspidal piece is concentrated in degree one, the resulting characteristic polynomial is quadratic and coincides with the Hecke polynomial of π_x. The commutation established above ensures that the polynomial is attached to the Galois action on the π-isotypic multiplicity space. □

Remark 14.4. The theorem is one of the places where the original paper is genuinely deep. What matters for the logic of the proof is that the cohomological description is sufficiently precise to compare these two commuting correspondences and to read local parameters on the same two-dimensional multiplicity space.

15. Shtukas and the completion of the Langlands picture

The preceding sections explain Drinfeld's original modular-curve argument. Shtukas reveal why that argument is not an isolated miracle. They provide a geometry in which modifications, Frobenius, Hecke correspondences, and global Galois parameters coexist from the outset. For GL_2 with one pole fixed at ∞, elliptic sheaves are already a disguised shtuka theory. Once one allows several legs and works on higher-dimensional moduli stacks, the same geometry becomes the engine behind the full function-field Langlands correspondence.

15.1. Rank-two shtukas with two legs

Definition 15.1. Let S be an $\mathbb{F}_q$-scheme. A rank-two shtuka over S with zero $\alpha : S \to X$ and pole $\beta : S \to X$ consists of a vector bundle $\mathcal{E}$ of rank 2 on $X \times S$ together with an isomorphism

$$\phi : \mathcal{E}\big|_{(X\times S)\setminus(\Gamma_\alpha\cup\Gamma_\beta)} \xrightarrow{\sim} (\mathrm{Id}_X \times \mathrm{Fr}_S)^*\mathcal{E}\big|_{(X\times S)\setminus(\Gamma_\alpha\cup\Gamma_\beta)}$$

whose relative position at α is an elementary upper modification and at β is an elementary lower modification.

Away from the two graphs one may think of ϕ as a Frobenius descent datum. The two legs measure the places where this datum fails to extend. The zero and pole play completely symmetric roles until one freezes one of them at ∞ and imposes a periodicity normalization. That specialization brings us back to elliptic sheaves.

Proposition 15.2. *Fix the pole at $\beta = \infty$ and impose the standard periodicity condition at ∞. Then the groupoid of rank-two shtukas with one variable zero and fixed pole ∞ is equivalent to the groupoid of rank-two elliptic sheaves.*

Proof. Starting from an elliptic sheaf $(\mathcal{E}_i, j_i, t_i)$, set $\mathcal{E} = \mathcal{E}_0$. The inclusion $j_0 : \mathcal{E}_0 \hookrightarrow \mathcal{E}_1$ is an elementary modification at ∞, while

$$t_0 : (\mathrm{Id} \times \mathrm{Fr})^* \mathcal{E}_0 \hookrightarrow \mathcal{E}_1$$

is an elementary modification at the characteristic point. By composing j_0^{-1} with t_0 away from the two graphs, one obtains the shtuka isomorphism ϕ. Compatibility of the squares in the elliptic-sheaf definition ensures that the resulting modification data satisfy the shtuka cocycle condition.

Conversely, begin with a shtuka $(\mathcal{E}, \phi)$ with fixed pole at ∞. Repeatedly modify by the pole condition and its Frobenius translates to construct a sequence $\mathcal{E}_i$. The zero leg produces the maps t_i, and the pole modifications produce the maps j_i. Because the pole is fixed at ∞, after two steps one has twisted by $\mathcal{O}(\infty)$, giving the required periodicity in rank two. The two constructions are inverse to one another up to the evident canonical isomorphisms. □

Remark 15.3. This equivalence explains why Drinfeld could work with elliptic modules and elliptic sheaves without changing the underlying geometry. The language of shtukas becomes essential only when the number of legs increases and when one wants to organize Hecke operators and Frobenius actions simultaneously on a moduli stack.

15.2. Partial Frobenius, Hecke correspondences, and boundary

Shtuka stacks have two families of correspondences built into their definition.

1. *Partial Frobenius*: one may apply Frobenius to one leg while keeping the other fixed.

2. *Hecke correspondences*: one may alter the bundle at an auxiliary finite place while keeping the legs unchanged.

The force of the theory is that these operations commute on the nose at the level of moduli problems and hence on cohomology.

Proposition 15.4. *Let* Sht_2 *denote the rank-two shtuka stack with two legs and fixed level away from the legs. Then for every finite place x away from the legs and the level, the Hecke correspondence T_x and the partial Frobenius operators* $\mathrm{Fr}_\alpha, \mathrm{Fr}_\beta$ *satisfy canonical commutative squares on* Sht_2.

$$\begin{tikzcd}
\mathrm{Sht}_2 \arrow[r, "T_x"] \arrow[d, "\mathrm{Fr}_\alpha"'] & \mathrm{Sht}_2 \arrow[d, "\mathrm{Fr}_\alpha"] & \mathrm{Sht}_2 \arrow[r, "T_x"] \arrow[d, "\mathrm{Fr}_\beta"'] & \mathrm{Sht}_2 \arrow[d, "\mathrm{Fr}_\beta"] \\
\mathrm{Sht}_2 \arrow[r, "T_x"'] & \mathrm{Sht}_2 & \mathrm{Sht}_2 \arrow[r, "T_x"'] & \mathrm{Sht}_2 .
\end{tikzcd}$$

Proof. A point of the shtuka stack is a vector bundle together with modification data at the two legs. Applying T_x changes the bundle by an elementary modification at the unrelated place x; it leaves the legs and their graphs untouched. Applying a partial Frobenius changes one leg by pullback under Frobenius on the base, again without touching the auxiliary place x. Since the three operations affect pairwise disjoint pieces of the data, they commute already before passing to isomorphism classes. The commutative diagrams are therefore tautological from the moduli description. □

The second issue is compactification. Shtuka stacks are typically nonproper. Their boundary consists of configurations in which the bundle becomes unstable or splits into lower-rank pieces. For rank two this boundary is exactly where Eisenstein contributions originate.

Theorem 15.5 (Essential versus negligible cohomology, rank two)**.** *After a suitable compactification and truncation by Harder–Narasimhan polygons, the compactly supported cohomology of the rank-two shtuka stack decomposes into an* essential *part and a* negligible *part. The negligible part is generated by rank-one data and contributes only Eisenstein series; the essential part is the one carrying cuspidal automorphic representations.*

Proof. The Harder–Narasimhan truncation cuts the stack into a quasi-compact open substack on which the degrees of destabilizing subbundles are bounded. The complement is stratified by parabolic reductions. In rank two every proper parabolic is the Borel, so the boundary strata are built from line bundles and extension classes between them. Their cohomology is therefore induced from rank-one automorphic data and produces only Eisenstein terms.

The compactly supported cohomology of the truncated open is finite dimensional and stable under Hecke correspondences and partial Frobenius. Passing to the direct limit over the truncation parameter isolates a maximal quotient on which no rank-one induced piece survives; this is the essential part. Because the boundary has been described in terms of parabolic reductions, the quotient is exactly the cuspidal cohomology. □

Remark 15.6. In Drinfeld's original modular-curve setting this distinction is far more concrete. The boundary consists of finitely many cusps, and compact support simply means functions vanish there. In higher-rank shtuka geometry one needs the Harder–Narasimhan truncation because the boundary is stacky and highly noncompact, but the representation-theoretic meaning is the same.

15.3. From Drinfeld to Laurent Lafforgue and Vincent Lafforgue

The passage from Drinfeld's theorem to the modern global function-field Langlands correspondence is not a change of philosophy. It is the same philosophy carried out on more elaborate shtuka stacks.

Theorem 15.7 (Laurent Lafforgue, structural form)**.** *For every $n \geq 1$, cuspidal automorphic representations of* $\mathrm{GL}_n(\mathbb{A}_K)$ *are naturally in bijection with irreducible n-dimensional ℓ-adic Galois representations of* $\mathrm{Gal}(K^{\mathrm{sep}}/K)$*, with compatibility of local unramified parameters at almost all places.*

Proof sketch. The moduli stack of rank-n shtukas with sufficiently many legs carries commuting actions of Hecke correspondences and partial Frobenius operators. After compactification and truncation, one studies the compactly supported cohomology of these stacks and applies the Grothendieck–Lefschetz trace formula to correspondences built from Hecke operators and Frobenius. The trace identities are compared with the Arthur–Selberg trace formula on the automorphic side.

The geometric input needed to control the boundary is vastly subtler than in rank two: one proves that the boundary contributes only automorphic data coming from proper Levi subgroups, hence lower rank. This allows an induction on n. The essential cohomology then realizes precisely the cuspidal spectrum of GL_n, and the eigenvalues of partial Frobenius produce the required n-dimensional Galois representations. □

Theorem 15.8 (Vincent Lafforgue, structural form)**.** *Let G be a connected reductive group over K. The cuspidal automorphic spectrum of $G(\mathbb{A}_K)$ admits a canonical decomposition indexed by semisimple global Langlands parameters*

$$\sigma : \mathrm{Gal}(K^{\mathrm{sep}}/K) \longrightarrow {}^L G(\overline{\mathbb{Q}}_\ell),$$

constructed by excursion operators on the cohomology of stacks of shtukas with many legs.

Proof sketch. One no longer tries to extract a Galois representation directly from a two-dimensional multiplicity space. Instead, one builds a large commutative algebra of endomorphisms of cuspidal automorphic forms, the *excursion algebra*. Its generators come from creating several legs on a shtuka, transporting them by Galois elements through partial Frobenius, and then annihilating them. The resulting operators depend functorially on representations of the dual group and on tuples of Galois elements.

The formal relations among excursion operators match exactly the tensor relations defining a semisimple L-parameter. Simultaneously, the Satake isomorphism identifies the Hecke operators with characters of the dual group at unramified places. Diagonalizing the excursion algebra therefore decomposes cuspidal automorphic forms into generalized eigenspaces indexed by semisimple global parameters. This provides the global parameterization for all reductive groups over function fields. □

Remark 15.9. Drinfeld's modular curve should be remembered as the smallest complete laboratory in which the whole mechanism can still be drawn by hand. Once one understands why elliptic sheaves are fixed-pole shtukas and why compactly supported cohomology removes the boundary, Laurent and Vincent Lafforgue's theorems stop looking like unrelated later miracles. They are the higher-rank and many-legged continuation of the same geometry.

16. Drinfeld's reciprocity law

We can now assemble the argument.

Theorem 16.1 (Drinfeld)**.** *For every irreducible cuspidal automorphic representation π of $\mathrm{GL}_2(\mathbb{A}_K)$ with $\pi_\infty \simeq \mathrm{St}$, the multiplicity space $\sigma(\pi)_\ell$ is a 2-dimensional irreducible ℓ-adic representation of $\mathrm{Gal}(K^{\mathrm{sep}}/K)$. The rule*

$$\pi \longmapsto \sigma(\pi)_\ell$$

induces a bijection between such automorphic representations and the corresponding irreducible Galois representations. The bijection is characterized by equality of local L-factors at every unramified finite place.

Proof. The proof breaks into six steps.

Step 1: geometric realization. By the moduli interpretation and adelic uniformization, the rank-two modular curve with level I is analytically

$$X_I = \mathrm{GL}_2(K)\backslash(\Omega^2 \times \mathrm{GL}_2(\mathbb{A}^\infty)/U(I)).$$

Step 2: cohomological calculation. The compactly supported cohomology $H^1_c(X_I, \mathbb{Q}_\ell)$ is identified with adelic harmonic cochains on the Bruhat–Tits tree, hence with the special representation at ∞.

Step 3: automorphic decomposition. Consequently,

$$H^1_c(\varinjlim_I X_I, \mathbb{Q}_\ell) \cong \bigoplus_{\substack{\pi \text{ cuspidal} \\ \pi_\infty \simeq \mathrm{St}}} \pi^\infty \otimes \sigma(\pi)_\ell,$$

where the multiplicity spaces $\sigma(\pi)_\ell$ inherit continuous Galois actions.

Step 4: local comparison away from ramification. At every unramified finite place x, the Frobenius polynomial on $\sigma(\pi)_\ell$ equals the Hecke polynomial of π_x. Therefore the two sides have the same unramified local parameters and the same local L-factors.

Step 5: irreducibility and dimension. Because the relevant cohomology is concentrated in degree one and the Hecke polynomial is quadratic, $\sigma(\pi)_\ell$ is two-dimensional. Irreducibility follows from the cuspidality of π and the purity of the cohomological realization: a reducible Galois representation would force an Eisenstein contribution, which does not occur in the compactly supported cuspidal cohomology.

Step 6: bijectivity. Injectivity follows from the equality of local Hecke/Frobenius polynomials at almost all places together with strong multiplicity one for GL_2. Surjectivity is obtained by reversing the argument: every Galois representation satisfying the same local conditions contributes to the same cohomology, hence to an automorphic isotypic component with the prescribed infinite type. Since the local L-factors match at almost all places, the correspondence is inverse on both sides.

This proves the theorem. □

Corollary 16.2. *The central character of π equals the determinant of $\sigma(\pi)_\ell$ under global class field theory, and twisting π by a Hecke character corresponds to twisting $\sigma(\pi)_\ell$ by the associated Galois character.*

Proof. Both assertions are already true at the level of the commuting Hecke and Galois actions on cohomology. The determinant of Frobenius on the multiplicity space is the constant term of the local Hecke polynomial, which is the local value of the central character times the correct power of q_x. Class field theory translates this identity into a global statement. Twisting behaves functorially on both automorphic forms and on the cohomological correspondences, so the compatibility follows place by place. □

Remark 16.3 (A structural reading of the proof)**.** The proof is layered, but it is not fragmented. Formal $\mathcal{O}$-modules isolate the local Frobenius-linear algebra. Drinfeld modules globalize that algebra over the affine curve. Level structures rigidify the moduli problem and create honest algebraic varieties. Elliptic sheaves move the same data to the complete curve, where vector bundles and Riemann–Roch control the pole filtration. Rigid uniformization translates the modular curve into an analytic quotient of the upper half-plane. The Bruhat–Tits tree converts that analytic geometry into a combinatorial model. Harmonic cochains identify the resulting cohomology with the special representation. The automorphic decomposition and the Galois action are then two ways of reading the same cohomology space. At the end one has not compared unrelated worlds; one has exhibited automorphic forms and Galois representations as two coordinate systems on one geometric object.

17. Final remarks

The function-field case solved by Drinfeld is historically the first large-scale success of the Langlands program beyond abelian class field theory. Its enduring importance is not only that it proves a correspondence, but that it exhibits a method: modular geometry, analytic uniformization, cohomology, and automorphic–Galois comparison belong to one uninterrupted argument. The geometric dictionary with elliptic sheaves is especially revealing. It shows that the algebra of Frobenius polynomials is the affine shadow of a theory of vector bundles with periodic Frobenius modifications on the complete curve. Once this is understood, Riemann–Roch, moduli, and cohomology cease to be external tools and become native parts of the same story.

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